Notes: Jiang, Stanford, & Xie (JFE, 2012)

Does it matter who pays for bond ratings? Historical evidence

Contents

1 Big picture

- **Question.** Does the issuer-pay compensation model cause credit rating agencies to assign higher ratings than they would under an investor-pay model?
- **Historical shock.** Moody's began charging corporate bond issuers in October 1970, while S&P continued to use investor-pay for corporate bonds until July 1974. This creates a period in which the two agencies used different revenue models for the same market.
- **Research design.** The paper compares S&P and Moody's ratings for the same newly issued corporate bond before and after S&P adopts issuer-pay. Moody's rating for the same bond is used as a benchmark.
- **Main finding.** Before July 1974, when Moody's charged issuers and S&P charged investors, S&P ratings were lower than Moody's ratings. After S&P adopts issuer-pay, the average gap disappears.
- **Mechanism test.** The increase in S&P ratings is concentrated where issuer-pay conflicts should be strongest: bonds expected to generate larger rating fees and bonds with lower credit quality within a Moody's rating category.
- **Magnitude.** The rating increase is about 20% of a broad rating grade. The paper translates this into roughly 10 basis points of yield-spread reduction, or about $51,000 per year in interest savings for the median issue in 1974 dollars.
- **Contribution.** The paper provides historical difference-in-differences evidence that the issuer-pay model can inflate ratings, complementing more recent evidence from structured finance and rating-agency competition.
- **Policy relevance.** The results speak to debates over whether ratings should be paid for by issuers or investors, and whether fee disclosure and stronger governance could reduce conflicts.

2 Data and measurement

2.1 Why the 1970s provide useful variation

- Ratings originally used an investor-pay subscription model: investors bought the rating manuals and reports.
- The growth of corporate and municipal debt markets increased demand for ratings in the late 1960s and early 1970s.
- Photocopying and fax technology worsened the free-rider problem, making it difficult for agencies to recover costs from subscribers.
- S&P began charging municipal issuers in 1968. Moody's began charging municipal issuers in 1970 and corporate issuers in October 1970. S&P began charging corporate issuers only in July 1974.

Interpretation: The key empirical window is 1971 through June 1974: Moody's is issuer-pay and S&P is investor-pay. After July 1974, both are issuer-pay. This staggered timing lets the authors compare two rating agencies for the same bond under different fee regimes.

2.2 Sample construction

- The sample contains corporate bonds issued between 1971 and 1978.
- Bonds must have both S&P and Moody's ratings.
- Ratings and bond characteristics come from Securities Data Corporation's Global New Issues database.

- The authors verify or supplement ratings with Moody's Default Risk Database, S&P's Credit Ratings Database, Moody's Bond Survey, Bond Outlook, and the Fixed Income Investor.
- Firm characteristics come from Compustat and CRSP.
- The final sample contains 797 bonds issued by 359 firms.
- More than 90% of the sample bonds are senior bonds, and most are investment grade.

Interpretation: The design compares ratings on the same bond, so bond-level unobserved credit quality is partly controlled for by construction. The cross-agency comparison is the central strength of the sample.

2.3 Rating transformations

- Ratings are transformed into numerical values from 1 to 7, where higher numbers mean better credit quality.
- S&P begins using plus/minus modifiers in 1974, while Moody's does not use the finer notch system until later.
- To make the agencies comparable, the paper collapses S&P plus/minus ratings into broad categories matching Moody's broad categories.

The key rating variables are:

$$\begin{aligned} \text{Sprating}_i &= \text{numerical S\&P rating for bond } i, && (1)\\ \text{Mdrating}_i &= \text{numerical Moody's rating for bond } i, && (2)\\ \text{Ratingdiff}_i &= \text{Sprating}_i - \text{Mdrating}_i. && (3) \end{aligned}$$

Interpretation: If Ratingdiff < 0, S&P gives the bond a lower rating than Moody's. If Ratingdiff $= 0$, the two agencies agree. If Ratingdiff > 0, S&P gives a higher rating.

2.4 Treatment period and conflict proxies

- $\text{Post74}_i = 1$ if the bond is issued from July 1974 through 1978, when S&P charges issuers for corporate bond ratings.
- $\text{Post74}_i = 0$ if the bond is issued before July 1974, when S&P still charges investors.
- $\text{Highfee}_i = 1$ if the bond issue size is above the sample median and the issuer's issue frequency is above the sample median.
- $\text{Lowcredit}_i = 1$ if the issuer's operating profit margin is below the median within the same year and Moody's rating category.

Interpretation: Highfee captures fee importance and repeat-business value. Lowcredit captures issuers that may have stronger incentives to pressure S&P to move them upward within or across broad rating categories.

2.5 Control variables

The main controls are firm and issue characteristics that can affect ratings:

- size: log total assets;
- leverage: debt divided by market value of total assets;
- margin: operating income before depreciation divided by total assets;

- stock return volatility: standard deviation of daily stock returns in the issue year;
- issue size: log borrowing amount;
- maturity: log years to maturity;
- seniority: indicator for senior claims.

Interpretation: The regressions also interact these controls with Post74, which allows rating standards and the weight on observables to change after 1974.

3 Models and empirical specifications

3.1 Baseline difference-in-differences model

The baseline model is:

$$Ratings_i = a_0 + a_1 \text{Post74}_i + b' Controls_i + c'(Controls_i \times \text{Post74}_i) + \varepsilon_i, \tag{4}$$

where the dependent variable can be Ratingdiff_i, Sprating_i, or Mdrating_i.

When $Ratings_i = \text{Ratingdiff}_i$, the coefficient a_1 measures the change in S&P's rating relative to Moody's after S&P switches to issuer-pay.

Interpretation: A positive a_1 means S&P becomes more favorable relative to Moody's after S&P begins charging issuers. Because Moody's rates the same bond, the design controls for much of the bond's credit quality.

3.2 Cross-sectional conflict-of-interest model

The conflict model is:

$$\text{Ratingdiff}_i = a_0 + a_1 \text{Post74}_i + a_2 C_i + a_3(\text{Post74}_i \times C_i) \tag{5}$$
$$+ b' Controls_i + c'(Controls_i \times \text{Post74}_i) + \varepsilon_i, \tag{6}$$

where C_i is either Highfee_i or Lowcredit_i.

- a_1 measures the post-1974 change for low-conflict bonds.
- a_3 measures whether S&P becomes more favorable for high-conflict bonds.
- The post-1974 change for high-conflict bonds is $a_1 + a_3$.

Interpretation: A significant positive a_3 is more diagnostic than a positive average post-1974 change. It shows that the rating increase occurs precisely where issuer-pay incentives should matter.

3.3 Yield-spread translation

To interpret the economic magnitude, the paper estimates how rating levels map into yield spreads. An improvement of one broad rating grade is associated with about a 48-basis-point reduction in yield spread. The estimated rating increase of roughly 20% of a grade therefore implies about a 10-basis-point spread reduction.

Interpretation: The rating effect is small in rating units but economically meaningful for bond issuers, especially for large or frequent borrowers.

4 Identification and empirical design

4.1 What identifies the effect

- The treated agency is S&P; the benchmark agency is Moody's.
- The pre period is 1971 through June 1974, when Moody's charges issuers and S&P charges investors.
- The post period is July 1974 through 1978, when both agencies charge issuers.
- The outcome is the same bond's S&P rating minus its Moody's rating.

Interpretation: The identifying idea is that any bond-specific credit quality should affect both agencies' ratings for that bond. A change in S&P relative to Moody's around S&P's fee-model switch is therefore informative about the fee model.

4.2 Why cross-sectional tests matter

- A simple before-after change could be caused by unrelated changes in S&P rating methodology.
- The paper therefore tests whether the post-1974 S&P increase is larger for bonds where issuer-pay conflicts are stronger.
- The strongest evidence is that the interaction terms Post74 $\times$ Highfee and Post74 $\times$ Lowcredit are positive and significant.

Interpretation: If S&P had simply improved its information or changed methodology uniformly, the increase should not be concentrated among high-fee and low-credit bonds.

4.3 Alternative explanations addressed

- **S&P plus/minus ratings:** The paper collapses plus/minus notches into Moody's broad categories and checks that inferences are robust to deleting plus/minus observations.
- **More information from issuers:** If issuer-pay simply gave S&P more information, S&P ratings should become more strongly related to yields and less dispersed. The paper finds some evidence of more information, but it cannot explain the concentration of rating increases among high-conflict bonds.
- **Subordinated bonds:** The results hold when the sample is restricted to senior bonds only.
- **Econometric specification:** The results are robust to ordered logit, clustering within firms, and alternative windows such as 1972-1977 and 1973-1976.

Interpretation: The robustness section strengthens the inference that the main pattern is not due to sample composition, S&P's finer ratings, or a simple information improvement.

4.4 Limits of the design

- The paper does not observe the full private negotiation process between issuers and rating agencies.
- It does not directly estimate whether bond investors fully understood and priced the post-1974 S&P rating increase.
- It compares S&P with Moody's, not with an objective default benchmark.
- The historical setting differs from modern structured finance, where fee levels and rating-agency business lines are much larger.

Interpretation: The paper's strongest claim is that the issuer-pay model is associated with higher ratings in this historical setting, especially where conflicts are larger. It is not a full model of every modern rating agency conflict.

5 Tables and figures

5.1 Figure 1 and Table 1: institutional timeline and rating scale

Read: Figure 1:

- S&P starts charging municipal bond issuers in January 1968.
- Moody's starts charging municipal issuers in May 1970 and corporate bond issuers in October 1970.
- The sample begins in 1971, after Moody's has already adopted issuer-pay for corporate bonds.
- S&P starts charging corporate bond issuers only in July 1974.

Read: Table 1:

- Ratings are mapped into seven broad categories, with 7 as the highest quality and 1 as the lowest.
- S&P plus/minus ratings are collapsed into Moody's broader categories.
- The sample includes 797 bonds with ratings from both agencies.
- The sample is concentrated in A and Aa categories, but includes bonds from B through Aaa/AAA.

Economic message: The timeline supplies the natural experiment; the rating transformation makes S&P and Moody's ratings comparable despite S&P's post-1974 plus/minus refinements.

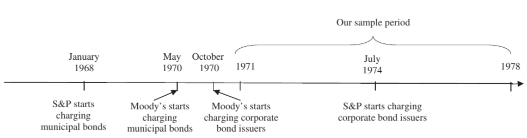

Fig. 1. Timeline of Moody's and S&P's change in revenue model. Change in revenue model refers to the change from investor-pay fee model to issuer-pay fee model.
Source: New York Times (January 27, 1968; February 3, 1968; and February 15, 1970); *Wall Street Journal* (February 15, 1968; October 1, 1970; and June 24, 1974).

(a) Revenue-model timeline.

Table 1
Distribution of new bond credit ratings in our sample period from 1971 to 1978.

Sprating and *Mdrating* are the numerical S&P and Moody's rating transformations. During our sample period, Moody's assigns only broad ratings (i.e., Aaa, Aa, A, Baa, Ba, B, Caa, Ca, and C). S&P begins assigning refined ratings (e.g., AA+ or AA−) in 1974. To consistently compare credit ratings both over time and across the two rating agencies, we ignore the refined S&P ratings and transform alphabetical ratings into numerical ratings based on Moody's broad rating categories. We believe this research design is conservative in terms of testing our hypotheses. We confirm that transforming the alphabetical ratings based on the refined ratings (i.e., from 1 to 17 rather than from 1 to 7) leaves our inferences unchanged. Our sample includes 797 new bonds issued between 1971 and 1978 that have both S&P and Moody's ratings and no credit-enhancement features.

S&P credit rating letter	Frequency (%)	Moody's credit rating letter	Frequency (%)	*Sprating* & *Mdrating*
AAA	6.02	Aaa	8.16	7
AA+	0.13			
AA	31.87	Aa	34.38	6
AA−	2.26			
A+	3.51			
A	34.38	A	38.27	5
A−	3.64			
BBB+	0.63			
BBB	10.16	Baa	12.55	4
BBB−	0.50			
BB+	0.25			
BB	2.13	Ba	2.76	3
BB−	0.63			
B+	0			
B	3.76	B	3.89	2
B−	0			
CCC	0.13	Caa/Ca/C	0	1

(b) Rating-to-number mapping.

5.2 Table 2: descriptive evidence on rating differences

Read:

- Panel A shows that before July 1974, S&P ratings are below Moody's ratings by about 0.11 broad rating grades on average.
- After July 1974, the average difference is approximately zero.
- Panel B shows the distribution of split ratings. Before July 1974, when ratings differ, S&P gives the lower rating in 40 of 51 split-rating cases, or 78%.
- After July 1974, S&P gives the lower rating in 43 of 85 split-rating cases, or 51%.
- Panel C shows the largest improvements in the Baa/Ba/Aaa zones, which are economically important because rating boundaries can affect investment constraints and regulation.
- Panel D shows that high-fee and low-credit bonds had much more negative S&P-Moody's gaps before 1974, and these gaps largely disappear after S&P adopts issuer-pay.
- Panel E shows that the largest changes occur for high-fee or low-credit bonds near the investment-grade boundary.

Economic message: The raw data already point to the paper's main result: S&P moves from being systematically more conservative than Moody's to being roughly equal once S&P is paid by issuers.

Table 2
Descriptive statistics.

The sample consists of 797 corporate bonds issued between 1971 and 1978 with a rating from both S&P and Moody's.

***, **, And * indicate it is significantly different from zero at $p < 0.01$, 0.05, and 0.10, respectively.

[1] In Panel C, the two-tailed p-values are from a t-test of whether the difference between S&P's ratings and Moody's ratings (*Ratingdif*) has changed significantly between the two sample periods.

[2] In Panel D, we use two proxies for the potential to generate large conflicts of interest. First, larger bond issuances (above the sample median) and bonds issued by firms that issue more frequently (more often than the sample median) are likely to generate higher fees for credit rating agencies. Second, issuers that have low credit quality (below the median profit margin within each Moody's rating category each year) have more incentives to push for the same ratings from S&P.

Panel A: Average rating differences for new bonds issued between 1971 and 1978

Year	Mean *Sprating*	Mean *Mdrating*	Mean *Ratingdif* (*Sprating* – *Mdrating*)	No. of obs. (Total = 797; N = 136 for *Sprating* ≠ *Mdrating*)
1971	5.65	5.69	−0.05	65
1972	5.10	5.19	−0.10**	84
1973	5.32	5.49	−0.17**	53
01/74–06/74	5.33	5.48	−0.15**	60
07/74–12/74	5.39	5.49	−0.10***	69
1975	5.26	5.26	−0.01	164
1976	5.14	5.13	0.01	123
1977	5.14	5.11	0.03	93
1978	4.43	4.40	0.03	86
Summary:				
1971–06/74	5.33	5.44	−0.11***	262
07/74–1978	5.093	5.095	−0.002	535

Panel B: Distributions of new bond ratings: from 1971 to June 1974 and from July 1974 to 1978 Observations on the diagonal receive the same rating from both agencies. Observations below (above) the diagonal receive a higher (lower) rating from Moody's.

	From 1971 to June 1974 S&P rating					
Moody's rating	B	BB	BBB	A	AA	AAA
B	3					
Ba	**3**	5				
Baa		**1**	23	**2**		
A			**4**	72	**8**	
Aa				**20**	94	**1**
Aaa					**12**	14

Of the 51 bonds receiving split ratings, S&P gives 40 bonds (i.e., 78%) a lower rating than Moody's does.

	From July 1974 to 1978 S&P rating					
Moody's rating	B	BB	BBB	A	AA	AAA
B	23	**4**				
Ba	**1**	12	**1**			
Baa		**2**	55	**17**		
A			**7**	196	**18**	
Aa				**24**	133	**2**
Aaa					**8**	31

Of the 85 bonds with split ratings, S&P gives 43 bonds (i.e., 51%) a lower rating than Moody's does. One bond rated CCC by S&P and B by Moody's is omitted from this panel.

Panel C: Rating differences across Moody's rating grades

	Observations		*Ratingdif* (*Sprating*–*Mdrating*)		Test of difference in pre- v. post-1974	
Moody's rating	Pre-July 1974	Post-July 1974	Pre-July 1974	Post-July 1974	Post – Pre	Two tailed p-values[1]
B	3	28	−0.00	0.11	0.11	0.18
Ba	8	14	−0.38*	−0.00	0.38	0.07
Baa	26	74	0.04	0.20***	0.16	0.06
A	84	221	0.05	0.05**	0.00	0.96
Aa	115	159	−0.17***	−0.19***	−0.02	0.57
Aaa	26	39	−0.46***	−0.21	0.25	0.03
Total	262	535	−0.11***	−0.002	0.11	0.00

Figure 2: Table 2, panels A-C: average differences and rating matrices.

Table 2 (*continued*)

Panel D: Distributions of rating differences for bonds with a low or high potential for conflicts of interest[2]

	S&P > Moody's	S&P = Moody's	S&P < Moody's	Total	*Ratingdif* (*Sprating–Mdrating*)
For bonds with low expected fees (borrow less and borrow less frequently)					
Pre-74	9 (5%)	138 (83%)	19 (11%)	166	−0.06*
Post-74	24 (7%)	287 (86%)	24 (7%)	335	0.00
For bonds with high expected fees (borrow more and borrow more frequently)					
Pre-74	2 (2%)	73 (76%)	21 (22%)	96	−0.20***
Post-74	18 (9%)	163 (82%)	19 (10%)	200	−0.005
For bonds that have higher creditworthiness (above—median profit margins within each year and Moody's rating category)					
Pre-74	10 (8%)	112 (84%)	11 (8%)	133	−0.01
Post-74	11 (4%)	243 (89%)	20 (7%)	274	−0.03
For bonds that have lower creditworthiness (below—median profit margins within each year and Moody's rating category)					
Pre-74	1 (1%)	99 (77%)	29 (22%)	129	−0.22***
Post-74	31 (12%)	207 (79%)	23 (9%)	261	0.03

Panel E: Is the impact of conflicts of interest more pronounced when a bond is close to investment grade? Rating difference (S&P - Moody's) is reported in each cell with the number of observations in parentheses.

Using high expected fees (borrow more and borrow more frequently) to proxy for conflicts of interest				
		Pre-74	Post-74	Change from pre- to post-74
Moody's rating is *not* Ba or Baa	Low expected fees	−0.06* (134)	−0.02 (269)	0.05
	High expected fees	−0.20*** (94)	−0.07** (178)	0.14**
		Pre-74	Post-74	Change from pre- to post-74
Moody's rating is Ba or Baa	Low expected fees	−0.06 (32)	0.06 (66)	0.12
	High expected fees	0.00 (2)	0.50*** (22)	0.50***
Using low creditworthiness (below-median profit margins within each year and Moody's rating category) to proxy for conflicts of interest				
		Pre-74	Post-74	Change from pre- to post-74
Moody's rating is *not* Ba or Baa	High creditworthiness	0.00 (114)	−0.05** (228)	−0.05
	Low creditworthiness	−0.24***(114)	−0.02 (219)	0.22***
		Pre-74	Post-74	Change from pre- to post-74
Moody's rating is Ba or Baa	High creditworthiness	−0.05 (19)	0.07 (46)	0.12
	Low creditworthiness	−0.07 (15)	0.29*** (42)	0.35***

Figure 3: Table 2, panels D-E: conflict proxies and investment-grade boundary evidence.

5.3 Tables 3 and 4: baseline regressions and conflict interactions

Read: Table 3:

- Column (1) estimates that after July 1974, S&P's rating relative to Moody's rises by 0.11 rating grades without controls.
- Column (2) estimates a 0.20 rating-grade increase with controls, with a one-tailed p-value of 0.08.
- Column (3) shows S&P's own average rating rises by 0.39 rating grades after July 1974.
- Column (4) shows Moody's rating does not significantly change after July 1974.
- These results imply that the disappearance of the Moody's-S&P gap is driven primarily by S&P increasing its ratings.

Read: Table 4:

- The Post74 × Highfee coefficient is positive and significant: 0.13 without controls and 0.19 with controls.
- The Post74 × Lowcredit coefficient is also positive and significant: 0.27 without controls and 0.22 with controls.
- The baseline Post74 coefficient is not significant in controlled specifications, meaning low-conflict bonds do not experience a similar rating increase.
- F-tests show that pre-1974 gaps exist for high-conflict bonds, but after 1974 their S&P ratings are no longer significantly below Moody's.

Economic message: Table 3 shows the average rating increase. Table 4 shows that the increase is concentrated exactly where issuer-pay conflicts of interest should be strongest.

Table 3
Test for whether S&P increases its ratings relative to Moody's ratings after S&P adopts the issuer-pay model in July 1974.

The sample consists of 797 corporate bonds issued between 1971 and 1978 with a rating from both S&P and Moody's. One-tailed robust *p*-values after adjusting for heteroskedasticity are in parentheses for variables with predicted signs including intercept and *Post*74 in column 1, 2, and 3. Two-tailed robust *p*-values after adjusting for heteroskedasticity are in parentheses for all other variables. *Post*74 is an indicator variable that takes the value of one when a bond is issued between July 1974 and 1978, the period wherein S&P charges issuers for ratings, and zero otherwise. Control variables include firm- and bond-specific characteristics and their interactions with *Post*74. These variables are described in the Appendix.

$$Ratings(Ratingdif, Sprating, Mdrating) = a_0 + a_1{*}Post74 + b_i{*}Controls(Size, Leverage, Margin, Stdret, Issuesize, Maturity, \text{and } Senior) + c_i{*}Controls{*}Post74 + Errors \quad (1)$$

	Ratingdif (=*Sprating*–*Mdrating*)		*Sprating*	*Mdrating*
Dep. variable:	(1)	(2)	(3)	(4)
Intercept	–0.11	–0.16	3.21	3.37
	(0.00)	(0.11)	(0.00)	(0.00)
Post74	0.11	0.20	0.39	0.19
	(0.00)	(0.08)	(0.06)	(0.43)
Size		−0.00	0.11	0.11
		(1.00)	(0.32)	(0.32)
Post74∗Size		0.01	0.09	0.08
		(0.81)	(0.43)	(0.52)
Leverage		0.05	−0.23	−0.27
		(0.90)	(0.72)	(0.62)
Post74∗Leverage		−0.32	−1.10	−0.78
		(0.41)	(0.10)	(0.19)
Margin		3.66	6.73	3.07
		(0.03)	(0.01)	(0.20)
Post74∗Margin		−4.31	−7.19	−2.88
		(0.01)	(0.01)	(0.25)
Stdret		−1.88	−45.71	−43.83
		(0.72)	(0.00)	(0.00)
Post74∗Stdret		2.13	10.75	8.62
		(0.73)	(0.31)	(0.39)
Issuesize		−0.01	0.23	0.24
		(0.85)	(0.16)	(0.14)
Post74∗Issuesize		−0.02	−0.06	−0.04
		(0.81)	(0.74)	(0.84)
Maturity		0.06	0.44	0.38
		(0.56)	(0.04)	(0.15)
Post74∗Maturity		−0.08	−0.41	−0.33
		(0.51)	(0.07)	(0.22)
Senior		0.08	2.23	2.15
		(0.55)	(0.00)	(0.00)
Post74∗Senior		−0.12	−0.59	−0.47
		(0.41)	(0.02)	(0.05)
With controls	No	Yes	Yes	Yes
Adjusted R^2 (%)	1	4	59	56

(a) Table 3: baseline regressions.

Table 4
Test for whether S&P increases its ratings for bonds that have greater conflicts of interest more than for other bonds after S&P adopts the issuer-fee model in July 1974.

The sample consists of 797 bonds issued between 1971 and 1978 with a rating from both S&P and Moody's. One-tailed robust *p*-values after adjusting for heteroskedasticity are in parentheses for the first six variables in the table and two-tailed *p*-values for all others. *Post*74 is an indicator variable that takes the value of one if a bond is issued between July 1974 and 1978, during which time S&P charges issuers for ratings, and zero otherwise. *Highfee* is an indicator variable that takes the value of one if the bond issue size is greater than the median issue size and the bond issuer's issue frequency is greater than the median issuer's issue frequency in our overall sample, and zero otherwise. *Lowcredit* is an indicator variable that takes the value of one if the issuer's margin is below the median within each year and Moody's rating category, and zero otherwise. Control variables include firm- and bond-specific characteristics and their interactions with *Post*74. These variables are described in the Appendix.

$$Ratingdif(= Sprating - Mdrating) = a_0 + a_1{*}Post74 + a_2{*}Highfee(Lowcredit) + a_3{*}Post74{*}Highfee(Lowcredit) + b_i{*}Controls + c_i{*}Controls{*}Post74 + Errors \quad (2)$$

	Ratingdif (= *Sprating*–*Mdrating*)			
Dependent variable:	(1)	(2)	(3)	(4)
Intercept	−0.06	−0.10	−0.01	−0.12
	(0.03)	(0.21)	(0.41)	(0.18)
Post74	0.06	0.12	−0.03	0.08
	(0.06)	(0.21)	(0.74)	(0.28)
Highfee	–0.14	–0.14		
	(0.01)	(0.04)		
Post74∗Highfee	0.13	0.19		
	(0.02)	(0.01)		
Lowcredit			−0.21	−0.11
			(0.00)	(0.08)
Post74∗Lowcredit			0.27	0.22
			(0.00)	(0.01)
Size		−0.02		−0.00
		(0.77)		(0.98)
Post74∗Size		0.03		0.01
		(0.62)		(0.86)
Leverage		0.15		0.03
		(0.69)		(0.94)
Post74∗Leverage		−0.47		−0.36
		(0.23)		(0.35)
Margin		3.55		2.40
		(0.03)		(0.25)
Post74∗Margin		−4.19		−2.51
		(0.01)		(0.24)
Stdret		−2.69		−2.08
		(0.61)		(0.71)
Post74∗Stdret		3.36		1.83
		(0.59)		(0.78)
Issuesize		0.07		−0.00
		(0.46)		(0.99)
Post74∗Issuesize		−0.13		−0.04
		(0.23)		(0.70)
Maturity		0.05		0.06
		(0.63)		(0.57)
Post74∗Maturity		−0.07		−0.07
		(0.57)		(0.52)
Senior		0.08		0.08
		(0.53)		(0.53)
Post74∗Senior		−0.12		−0.10
		(0.41)		(0.47)
Two-tailed *p*-value of *F*-tests for:				
S&P's average rating equal Moody's average rating for bonds coded as *Highfee* or *Lowcredit* **before** July 1974				
Intercept+*Highfee/Lowcredit*=0	0.00	0.07	0.00	0.09
S&P's average rating equal Moody's average rating for bonds coded as *Highfee* or *Lowcredit* **after** July 1974				
Intercept+*Post74*+*Highfee/Lowcredit*+*Post74∗Highfee/Lowcredit*=0	0.87	0.33	0.28	0.26
With controls	No	Yes	No	Yes
Adjusted R^2 (%)	2	4	4	4

(b) Table 4: high-fee and low-credit interactions.

5.4 Table 5 and Table A1: senior-bond robustness and variable definitions

Read: Table 5:

- The sample is restricted to 727 senior bonds.
- Panel A shows that the post-1974 S&P increase relative to Moody's remains positive.
- Panel B shows that the high-conflict interaction results survive: Post74 × Highfee and Post74 × Lowcredit remain positive and significant.
- This rules out the concern that subordinated bonds drive the main results.

Read: Table A1:

- Table A1 defines the key rating, treatment, conflict, and control variables.
- It clarifies that Highfee combines issue size and issuance frequency.
- It clarifies that Lowcredit is based on operating margin relative to peers in the same year and Moody's rating category.

Economic message: The senior-bond robustness is important because senior bonds are more homogeneous and constitute most of the sample. The variable definitions show that the conflict proxies are constructed to capture expected rating-fee value and issuer pressure within rating buckets.

Table 5
Robustness test using only senior bonds.

The sample consists of 727 senior bonds issued between 1971 and 1978 with a rating from both S&P and Moody's. One-tailed robust *p*-values after adjusting for heteroskedasticity are in parentheses. *Post*74 is an indicator variable that takes the value of one if a bond is issued between July 1974 and 1978, during which time S&P charges issuers for ratings, and zero otherwise. *Highfee* is an indicator variable that takes the value of one if the bond issue size is greater than the median issue size and the bond issuer's issue frequency is greater than the median issuer's issue frequency in our overall sample, and zero otherwise. *Lowcredit* is an indicator variable that takes the value of one if the issuer's margin is below the median within each year and Moody's rating category, and zero otherwise. Control variables include firm- and bond-specific characteristics and their interactions with *Post*74. All control variables are included but not reported for brevity. These variables are described in the Appendix.

Panel A: Test for whether S&P increases its ratings relative to Moody's after S&P adopts the issuer-pay model in July 1974 using only senior bonds

$$Ratings(Ratingdif, Sprating, Mdrating) = a_0 + a_1 * Post\ 74 + b_i * Controls(Size, Leverage, Margin, Stdret, Issuesize, Maturity, and\ Senior) + c_i * Controls * Post\ 74 + Errors \quad (1)$$

	Ratingdif (=Sprating–Mdrating)	
Dep. variable:	(1)	(2)
Intercept	−0.11	−0.07
	(0.00)	(0.04)
*Post*74	0.10	0.07
	(0.00)	(0.07)
No. of observations	727	727
With controls	No	Yes
Adjusted R^2 (%)	1	5

Panel B: Test for whether S&P increases its ratings for bonds that have greater conflicts of interest more than for other bonds after S&P adopts the issuer-fee model in July 1974 using only senior bonds

$$Ratingdif (= Sprating - Mdrating) = a_0 + a_1 * Post74 + a_2 * Highfee(Lowcredit) + a_3 * Post\ 74 * Highfee(Lowcredit) + b_i * Controls + c_i * Controls * Post\ 74 + Errors \quad (2)$$

	Ratingdif (= Sprating–Mdrating)			
Dep. variable:	(1)	(2)	(3)	(4)
Intercept	−0.05	−0.02	0.01	−0.03
	(0.06)	(0.38)	(0.59)	(0.23)
*Post*74	0.04	−0.02	−0.05	−0.03
	(0.15)	(0.63)	(0.89)	(0.69)
Highfee	−0.15	−0.13		
	(0.00)	(0.05)		
*Post74**Highfee*	0.16	0.22		
	(0.01)	(0.01)		
Lowcredit			−0.23	−0.11
			(0.00)	(0.10)
*Post74**Lowcredit*			0.30	0.22
			(0.00)	(0.02)
No. of observations	727	727	727	727
With controls	No	Yes	No	Yes
Adjusted R^2 (%)	2	5	4	5

Figure 5: Table 5: robustness using senior bonds only.

Table A1
The credit ratings and other bond-specific variables are extracted from Securities Data Corporation's Global New Issues database. The issuer characteristics are constructed from Compustat and CRSP.

Sprating	The numerical Standard & Poor's rating transformation, from 1 to 7 with greater numerical values corresponding to higher ratings.
Mdrating	The numerical Moody's Investor Service (i.e., Moody's) rating transformation, from 1 to 7 with greater numerical values corresponding to higher ratings.
Ratingdif	The difference between Standard & Poor's credit rating and Moody's credit rating for a bond (*Ratingdif*=*Sprating*−*Mdrating*).
*Post*74	An indicator variable that takes the value of one if the bond is issued between July 1974 and 1978 and zero otherwise.
Highfee	An indicator variable that takes the value of one if the bond issue size is greater than the median issue size and the bond issuer's issue frequency is greater than the median issuer's issue frequency in our overall sample and zero otherwise.
Lowcredit	An indicator variable that takes the value of one if the issuer's operating profit margin is below the median within each year and Moody's rating category and zero otherwise.
Size	The natural log of total assets (Compustat data # 6) at fiscal year-end before a bond is issued.
Leverage	Long-term debt (Compustat data # 9) plus the current portion of long-term debt in current liabilities (Compustat data # 34) divided by the market value of total assets, calculated as the book value of assets (Compustat data # 6) minus the book value of equity (Compustat data # 216) plus the market value of equity (Compustat data #25*#199).
Margin	Operating income before depreciation (Compustat data #13) divided by total assets (Compustat data # 6) at fiscal year-end before a bond is issued.
Stdret	The standard deviation of daily stock return during the year of issuance (from CRSP).
Issuesize	The natural log of borrowing amount in millions of dollars.
Maturity	The natural log of years to maturity.
Senior	An indicator variable that takes the value of one if the bond is a senior claim and zero otherwise.

Figure 6: Table A1: variable definitions.

6 Short summary

- The paper studies whether issuer-pay ratings are higher than investor-pay ratings.
- The key historical setting is that Moody's adopted issuer-pay for corporate bonds in 1970, while S&P adopted issuer-pay in July 1974.
- The authors compare S&P and Moody's ratings for the same corporate bonds issued from 1971 to 1978.
- The dependent variable is often Sprating − Mdrating, so Moody's rating for the same bond serves as the benchmark.
- Before July 1974, S&P ratings are lower than Moody's ratings by about 0.11 broad rating grades.
- After S&P adopts issuer-pay, the average difference between S&P and Moody's disappears.
- The change is driven by S&P raising ratings, not by Moody's lowering ratings.
- S&P raises ratings more for bonds expected to generate larger fees and for bonds with lower credit quality within Moody's rating buckets.
- The increase is also stronger near the investment-grade boundary, where ratings have higher regulatory and investment-policy value.
- The results are robust to senior bonds only, ordered logit, clustering, alternative windows, and treatment of S&P plus/minus ratings.
- The paper concludes that the issuer-pay model creates conflicts of interest that can lead to higher ratings.

7 Conclusion

7.1 Core conclusion

Jiang, Stanford, and Xie provide historical evidence that who pays for bond ratings matters. When S&P switches from investor-pay to issuer-pay for corporate bonds, its ratings become more favorable relative to Moody's. The effect is not uniform; it is concentrated in bonds where issuer-pay conflicts are likely to be strongest.

7.2 Economic intuition

The issuer-pay model gives bond issuers bargaining power because rating agencies are paid by the firms they rate. If issuers can generate large fees or benefit substantially from a higher rating, they have stronger incentives to pressure the rating agency. The agency may respond by assigning more favorable ratings, especially when the bond is close to a valuable rating threshold.

7.3 Takeaway

The paper's broad message is that rating-agency compensation can affect ratings. Reputation and market discipline may limit abuse, but they do not eliminate conflicts. The clean historical comparison between S&P and Moody's shows that when the payer changes, rating behavior changes too.